Deepak Pandey

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Education

Graphic Era Hill University, Dehradun **2021-2025**
B.TECH/Computer Science and Engineering *8.53/10.00*

Inspiration Senior Secondary School, Kathgodam **2020**
C.B.S.E *88.4/100*

Oakland Public School, Lohaghat **2018**
C.B.S.E *96.2/100*

Projects

Personalized Healthcare Assistant Integrating AI/ML Models **January 2025 – March 2025**

- Built a real-time multimodal mental health assistant in Streamlit, integrating emotion detection through Logistic Regression (TF-IDF) for text, Convolutional Neural Networks (CNN) for facial emotion, and Support Vector Machine (SVM) with MFCC features for voice emotion analysis.
- Optimized models to achieve 85% accuracy while ensuring minimal resource usage, enabling efficient inference on standard hardware.
- Streamlined data pipelines for fast pre-processing (vectorization, MFCC extraction, grayscale resizing), reducing model inference time.
- Deployed compact models (joblib for text and SVM, .h5 for facial emotion model) to minimize load time and ensure seamless integration within the app.
- Integrated emotion-based chatbot interaction, suggesting tailored responses and medicines based on real-time text, voice, and facial emotion inputs.
- Designed a user-friendly UI that consolidates text, facial, and voice input for an accessible, single-page mental health assessment application.

Face Emotion Detection Using Deep Learning **October 2024 – November 2024**

- Developed a real-time facial emotion recognition system using deep learning (CNN, VGG16, ResNet50), achieving 90% accuracy with 25% faster training using ResNet50.
- Enabled efficient and scalable emotion detection for real-world applications like mental health monitoring and human-computer interaction.
- Integrated OpenCV for optimized image processing and Gradio for an interactive UI, ensuring both performance and usability.

Movie Recommendation System **June 2024 – August 2024**

- Developed a content-based recommendation engine using TF-IDF and cosine similarity, optimizing resource usage for fast, real-time movie suggestions.
- Efficiently preprocessed and engineered features (genres, cast, keywords) for scalable recommendations without reliance on user ratings.
- Integrated Streamlit for instant, interactive movie suggestions, improving user experience with minimal latency.
- Designed for scalability, ensuring the system can handle large datasets with high performance and low computational overhead.

Technical Skills

- **Languages:** C++, C, Python, SQL, Java
- **Technologies/Frameworks:** TensorFlow, Pandas, scikit-learn, Hugging Face, PyTorch, Linux, NLP, CNN.
- **Coursework:** Data Structures and Algorithms, Machine Learning, Object-Oriented Programming, Computer System Security, Database Management Systems, Mathematics, Software Engineering

Research Paper

A Study on Prediction of Cardiac Disease Using Random Forest

- Conducted research on cardiac disease prediction using Random Forest, achieving 85.25% accuracy.
- Evaluated ML models (KNN, Decision Tree, Naive Bayes, Gradient Boosting) for comparative performance.
- Applied data preprocessing, feature selection, and model evaluation (accuracy, precision, recall, F1-score).
- Leveraged Python, Pandas, NumPy, Scikit-learn, and Seaborn for data analysis and model development.
- Demonstrated model robustness and generalization on healthcare datasets with minimal overfitting. Published and presented paper: A Study on Prediction of Cardiac Disease Using Random Forest.

Field of Interest

- Software Engineering and Development, AI and Machine Learning, Front-End Development.

Achievements

- Won 24 hours Hack-a-thon organized by the Computer Science Department of our university.

- 150+ problems solved on Leetcode.

- Organized and hosted CODEV CLUB seminars focusing on coding, development, and the impact of emerging technologies as Management Head.